26. Friday, May 30. Relations (7.1), Equivalence relations (7.2).

Last Time: A function $f: A \to B$ is the same as a subset $f \subseteq A \times B$ with certain properties.

Definition 1: Let A and B be sets. A relation from A to B is a subset $R \subseteq A \times B$.

A relation from the set A to itself is called a relation on A. (subset $R \subseteq A \times A$)

Example: Let $A = \{1, 2, 5\}$, $B = \{0, 4, 7\}$. Define a relation R from A to B by

$(x, y) \in R$ if and only if $x - y$ is even.

Here

$A \times B = \{(1,0), (1,4), (1,7), (2,0), (2,4), (2,7), (5,0), (5,4), (5,7)\}$

$R = \{(1,7), (2,0), (2,4), (5,7)\} \subseteq A \times B$.

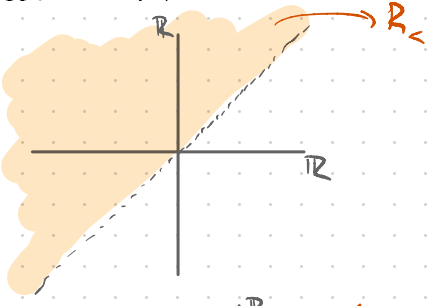
Example: ($A = B = \mathbb{Z}$) We define a relation R on $\mathbb{Z}$ by congruence modulo 5:

$(a, b) \in R$ if and only if $a \equiv b \pmod 5$

$R \geq \{(0,0), (0,5), (0,10), \dots, (2,102), (3,103) \dots, (3,-2)\}$

Example: Define a relation $R_<$ on $\mathbb{R}$ by "less than", i.e:

$$R_< = \{(x,y) \in \mathbb{R}^2 \mid x < y\} \subseteq \mathbb{R} \times \mathbb{R}.$$

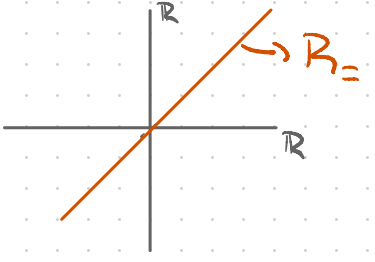


E.g. $R_< \supseteq \{(0,1), (3,100), (\pi, 100) \ldots\}$

But $(1,0) \notin R_<$, $(100,1) \in R_<$.

Another relation $R_=$ on $\mathbb{R}$

$$R_= = \{(x,y) \in \mathbb{R}^2 \mid x = y\}$$



Example: The "divides" relation on $\mathbb{Z}$

$$R_{div} = \{(m,n) \in \mathbb{Z} \times \mathbb{Z} \mid m \text{ divides } n\}$$

So $(3,3)$, $(3,6)$, $(5,10) \in R_{div}$ but $(6,3)$, $(7,5) \notin R_{div}$.

Notation: Let R be a relation from A to B. Let $x \in A$, $y \in B$. We write

$x R y$ to mean $(x,y) \in R$

$x \not R y$ to mean $(x,y) \notin R$

`$x ~R~ y$`

Sometimes we use a generic symbol instead of R, usually "$\sim$", so

$x \sim y$ means $(x,y) \in R$

$x \not\sim y$ means $(x,y) \notin R$

`$x \sim y$`

`$x \not\sim y$`

Example: in R_{div}, we write $3 \, R_{div} \, 6$ instead of $(3,6) \in R_{div}$

$5 \not R_{div} \, 7$ instead of $(5,7) \notin R_{div}$

Equivalence Relations (7.2)

Definition 2: Let A be a non-empty set. Let R be a relation on A.

We say that R is:

1) **Reflexive** if $\forall x \in A$, we have xRx (i.e. $(x,x) \in R$)

2) **Symmetric** if $\forall x,y \in A$, if xRy then yRx (i.e. $(x,y) \in R \Rightarrow (y,x) \in R$)

3) **Transitive** if $\forall x,y,z \in A$, if xRy and yRz, then xRz (i.e. $(x,y), (y,z) \in R \Rightarrow (x,z) \in R$)

Examples: • $R_<$ on $\mathbb{R}$. We write $x < y$ instead of $xR_<y$

Not reflexive ($0 \not< 0$) Not symmetric ($0<1$ but $1 \not< 0$)

It is transitive: if $x<y$ and $y<z$, then $y-x$ positive and $z-y$ positive.

Hence $z-y + y-x = z-x$ positive, so $x<z$.

- Equality relation $R_=$ on $\mathbb{R}$ satisfies all three (by axioms)
- What about divides?
- Congruence modulo 5?

Definition 3: A relation on a nonempty set A is called an **equivalence relation** if it is reflexive, symmetric and transitive.

Example: Let $A = \{$ cities/towns in USA$\}$. Define relation $\sim$ on A by

$x \sim y$ if and only if x is in the same state as y.

This is an equivalence relation.

Example: Fix $n \in \mathbb{N}$. We have the relation "congruence mod n" on $\mathbb{Z}$:

$a \sim b$ if and only if $a \equiv b \pmod{n}$

We proved this is an equivalence relation!

Equivalence classes (7.3)

Motivation: congruence modulo 3 on $\mathbb{Z}$

$$\ldots \equiv -6 \equiv -3 \equiv 0 \equiv 3 \equiv 6 \equiv 9 \equiv 12 \ldots \pmod 3$$
$$\ldots \equiv -5 \equiv -2 \equiv 1 \equiv 4 \equiv 7 \equiv 10 \equiv 13 \ldots \pmod 3$$
$$\ldots \equiv -4 \equiv -1 \equiv 2 \equiv 5 \equiv 8 \equiv 11 \equiv 14 \ldots \pmod 3$$

Each integer appears in exactly one of these three!!

Definition 4: Let $\sim$ be an equivalence relation on a set A. Let $a \in A$.

The **equivalence class of a** is the subset

$$[a] = \{ x \in A \mid x \sim a \}$$

Example: For "congruence mod 3" on $\mathbb{Z}$, we have

$$[0] = \{ x \in \mathbb{Z} \mid x \equiv 0 \pmod 3 \} = [3] = [6] = \cdots$$

$$[1] = \{ x \in \mathbb{Z} \mid x \equiv 1 \pmod 3 \} = [4] = [7] = \cdots$$

$$[2] = \{ x \in \mathbb{Z} \mid x \equiv 2 \pmod 3 \} = [5] = [8]$$

Example: For $A = \{$ cities/towns $\}$ and "same state relation"

$$[\text{Seattle}] = \{ \text{Seattle, Tacoma, Yakima, } \ldots \} = \{ \text{All cities/towns in WA} \}$$
$$= [\text{Yakima}]$$

$$[\text{Los Angeles}] = \{ \text{San Fco, San Diego, } \ldots \} = \{ \text{All cities/towns in CA} \}$$